S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

Activity No. 7

Aim: Apply the concept of network to configure the IPV4 for private and public IPs.

Name of Student:

Roll No.:

Semester/Year:

Academic Session:

Date of Performance:

Date of Submission:

AIM: Apply the concept of network to configure the IPV4 for private and public IPs.

OBJECTIVE / EXPECTED LEARNING OUTCOME:

The objectives and expected learning outcome of this practical are:

Make use of the concept of networking to configure the Router, Access point, and IPV4 for private and public IPs.

HARDWARE AND SOFTWARE REQUIREMENTS:

Hardware Requirement

- Processor: Dual Core
- RAM: 1GB
- Hard Disk Drive : > 80 GB

Software Requirement

- Networking Tool

THEORY:

STEPS FOR PRACTICAL:

Router setup steps

Step 1: **Decide where to place the router**

The best place for a wireless business router is in an open area of the workplace, as you'll benefit from even coverage. However, sometimes it's not easy to find a space out in the open because you must connect the router to a broadband gateway from your ISP (Internet service provider), which is usually attached to a cable near an outside wall.

Step 2: **Connect to the Internet**

Attach the router to a cable - or choose a mesh router

To solve the "long-distance" problem when connecting a router, you can use a CAT5e or CAT6 cable to connect the router to the ISP gateway's Ethernet port. Another option is to run Ethernet cables through the walls of your office to the chosen central location for the router.

Yet another option is to install a mesh network with a router. A mesh network allows you to place multiple Wi-Fi transmitters across your home or office, all on one network. Unlike extenders, which can be used with any wireless router, mesh networks require a router with this capability built-in.

No matter which option you choose, you'll use a basic Ethernet cable, plugged into the router's wide-area network (WAN) or Internet port. The Internet port is typically set apart from other ports by a different color.

Check the router's LED lights

Your router's LED lights tell you if you've successfully made an active Internet connection. If you don't see lights confirming such a connection, make sure you've plugged the cable into the correct port.

Test the connection with a device

Confirm that your router has a working connection by plugging a laptop computer into one of the device ports on the back of the router. If all goes well, you should be able to begin a wired connection, just as you did when confirming an active Internet connection.

Step 3: **Configure the wireless router gateway**

In some cases, ISPs offer customers gateways with built-in routers. In most cases, these combined devices are not built for business environments, nor do they have extra ports, security, and other options that allow you to add services and expand networks as the business grows.

If you have a gateway with an integrated router, you'll have to configure the gateway to disable the router and pass the WAN IP address—the unique Internet protocol address that the Internet provider assigns to your account—and all network traffic through to your new router.

If you don’t take this step, you may run into conflicts that prevent devices from working properly. You may need to contact your ISP for help with this step.

Step 4: **Connect gateway to router**

First, turn off the gateway. If there is already an Ethernet cable plugged into the gateway's local-area network (LAN) port, unplug the cable and plug it into your router's WAN port. Turn the gateway back on and wait a few minutes for it to boot up. Plug in the router's power supply and turn it on, again waiting a few minutes.

Step 5: **Use app or web dashboard**

The easiest way to continue with router setup is to use a mobile app if the router maker provided one. If there is no app, or you'd rather use the router's web-based dashboard, connect the router to a computer via an Ethernet cable.

You might find the router's IP address printed on the back of the device itself; if not, type 192.168.1.1, a common router address, into the browser search bar.

Step 6: **Create a username and password**

To configure the router, you'll need to log in, using its default admin name and password. You can usually find this information printed on the router itself, or in an accompanying user manual. Next, enter the required credentials. Once you're in, you should immediately create a new username and password. The defaults are usually something like "admin" and "password1234," which are obviously not secure—so make sure to change them at the first opportunity.

Step 7: **Update the router's firmware**

Your router may need an update of the "firmware," or software that operates it. Update it as soon as possible, since the new firmware might fix bugs or offer new security protections.

Some routers may download new firmware automatically, but many do not. You may need to check for updates through the app or the browser interface.

Step 8: **Create a Wi-Fi password**

Just as most routers come with preassigned admin usernames and passwords, most also come with preset Wi-Fi usernames and passwords. You’ll likely be prompted to change the Wi-Fi username and password, but even if you don't see such a prompt, plan to do so quickly.

Step 9: **Use auto-configuration tools where possible**

If your router is equipped with auto-install features, rely on them to help complete setup. For example, you should be able to use auto-configuration to manage IP addresses with the Dynamic Host Configuration Protocol (DHCP), which automatically assigns IP addresses to devices. You can always change these addresses later.

Step 10: **Set up security**

Many router manufactures provide security functionality to safeguard network and user privacy. You can login into the web dashboard and enabling added security features such as firewall, web filtering, and access controls to protect yourself from malicious traffic. You can also set up virtual private networks (VPNs) for privacy.

The purchase of new equipment for home or office always entails a minor potential inconvenience of setting it up. Of course, some people enjoy tinkering around the house, so installing new devices is nothing but a delight. However, there are also people who find it a chore, who want to get it over with as soon as possible, or who do not want to risk damaging the equipment.

LigoWave is here to make your access point setup experience simple and smooth. Below are some general guidelines on how to set up an access point and what is necessary for the setup.

Please note that these guidelines are of general nature and may differ depending on the device or system.

Necessities. There are several things that users will need in order to set up their access points:

- **an access point** – the main item on the list. Access points provide an additional location for end devices to connect to a local area network, thus expanding the LAN's area of coverage.
- **a switch or router** – devices necessary for the distribution and management of data transferred to, among, and from access points and end devices within a local area network.
- **Ethernet cables** – wires necessary for high-speed data transfer among access points and other devices.
- **an AC/DC adapter** (if supplied) – some access points may need a power adapter to function. However, most modern access points support Power over Ethernet (PoE), which eliminates the need for external power adapters.
- **a computer** – may be necessary for basic (or advanced) access point or network configuration.

Several other items may also be needed depending on the hardware and situation. This may include a modem, access point driver/suite installation discs, extension cords, wall mounts, and other elements.

Setup. Step 1. Deploy the Router/Switch. If there is no router or switch installed yet, do so:

1. Find a good spot for the device.

If it is a wireless router, place it in a location that would be appropriate, keeping the potential location of the access point in mind. This guarantees effective distribution of coverage and minimized interference levels.

If it is a switch or a wired router, then deploy it in a location where it would not interfere with any activities and the surrounding interior solutions.

2. Connect the router to a power outlet and switch it on. PoE-compatible devices will not require a power outlet—just an Ethernet cable. Also, some devices turn on automatically as soon as they are connected to a power source.

Step 2. Deploy the Access Point.

1. Find a good spot for the device.

If the local area network has other wireless devices, such as a router, place the access point in a location where its wireless signal would have minimum or no interference with other wireless devices within the LAN and where it would provide effective coverage with respect to the premises.

If the purpose of the access point is to introduce wireless connectivity to a wired LAN, then place it in a location where the wireless signal would cover the greatest possible area with the least number of obstacles and the least amount of interference.

2. If the device is powered by an AC/DC adapter, connect it to the access point and plug it into an electrical socket. If it is a PoE connection, proceed to the next step.

Step 3. Connect the Cables. There are several cables that the user will need to connect.

1. Use an Ethernet cable to connect the access point to the router. The cable should be inserted into a LAN port on the router and into the main Ethernet port on the access point. This introduces the access point to the router's created local area network.
2. Use an Ethernet cable to connect the router to the computer. The cable should be inserted into a LAN port on the router and into the Ethernet port on the computer (typically there is only one). This puts the computer onto the network and allows direct access point and LAN management.
3. If it is necessary to provide access to the Internet over the local area network, use an Ethernet cable to connect the router to the modem. Insert the cable into the main Ethernet port on the router and into the Ethernet port on the modem (typically there is only one).

Step 4. Configure. Modern technologies have practically done away with the inconvenience of manual LAN setup, let alone Internet access setup. Once everything is properly connected and the devices have acknowledged each other's presence on the network, the user is ready to go.

If any basic setup is necessary either for the access point or for the network in general, the computer or the user manual should provide instructions on what specific steps to take.

Users can also change optional and advanced router settings by connecting to the router via the computer using the IP address 192.168.0.1 or 192.168.1.1 (or the address noted in the manual).
Step 5. Viola! The setup is complete! Users can now enjoy expanded wireless coverage and improved wireless signal with the newly deployed access point!

Simplify Your Access Point Experience. LigoWave's Infinity series offers three generations of

universal wireless access points that enable users by delivering sheer power and straightforward simplicity.

The latest addition to the Infinity series redefines the meaning of "power" with its outstanding 720MHz processor and high-performance omnidirectional dual-band antennas. By making full use of the 3×3 MIMO technology, the NFT 3AC wireless access point delivers a staggering 1.75Gbps of maximum aggregate throughput—more than enough to satisfy any local area network need.

Simplicity is key to making an already great product even greater. This is why the Infinity series of access points are designed to bring simplicity to the next level. These devices are ready-to-use right out of the box and the process of setting up is practically non-existent. Simply plug it in and it is all set!

The Infinity series of access points is a professional solution achieved using simple means. It enables all of its users to do more with less through being both powerful and simple.

What is IP?

An IP stands for internet protocol. An IP address is assigned to each device connected to a network. Each device uses an IP address for communication. It also behaves as an identifier as this address is used to identify the device on a network. It defines the technical format of the packets. Mainly, both the networks, i.e., IP and TCP, are combined together, so together, they are referred to as a TCP/IP. It creates a virtual connection between the source and the destination.
We can also define an IP address as a numeric address assigned to each device on a network. An IP address is assigned to each device so that the device on a network can be identified uniquely. To facilitate the routing of packets, TCP/IP protocol uses a 32-bit logical address known as IPv4(Internet Protocol version 4).
An IP address consists of two parts, i.e., the first one is a network address, and the other one is a host address.
There are two types of IP addresses:

- IPv4
- IPv6

What is IPv4?

IPv4 is a version 4 of IP. It is a current version and the most commonly used IP address. It is a 32-bit address written in four numbers separated by a 'dot', i.e., periods. This address is unique for each device.
For example, **66.94.29.13**
The above example represents the IP address in which each group of numbers separated by periods is called an Octet. Each number in an octet is in the range from 0-255. This address can produce 4,294,967,296 possible unique addresses.
In today's computer network world, computers do not understand the IP addresses in the standard numeric format as the computers understand the numbers in binary form only. The binary number can be either 1 or 0. The IPv4 consists of four sets, and these sets represent the octet. The bits in each octet represent a number.
Each bit in an octet can be either 1 or 0. If the bit the 1, then the number it represents will count, and if the bit is 0, then the number it represents does not count.
Representation of 8-bit Octet

128	64	32	16	8	4	2	1

The above representation shows the structure of 8 octets.

Now, we will see how to obtain the binary representation of the above IP address, i.e., 66.94.29.13

Step 1: First, we find the binary number of 66.

128	64	32	16	8	4	2	1
0	1	0	0	0	0	1	0

To obtain 66, we put 1 under 64 and 2 as the sum of 64 and 2 is equal to 66 (64+2=66), and the remaining bits will be zero, as shown above. Therefore, the binary bit version of 66 is 01000010.

Step 2: Now, we calculate the binary number of 94.

128	64	32	16	8	4	2	1
0	1	0	1	1	1	1	0

To obtain 94, we put 1 under 64, 16, 8, 4, and 2 as the sum of these numbers is equal to 94, and the remaining bits will be zero. Therefore, the binary bit version of 94 is 01011110.

Step 3: The next number is 29.

128	64	32	16	8	4	2	1
0	0	0	1	1	1	0	0

To obtain 29, we put 1 under 16, 8, 4, and 1 as the sum of these numbers is equal to 29, and the remaining bits will be zero. Therefore, the binary bit version of 29 is 00011101.

Step 4: The last number is 13.

128	64	32	16	8	4	2	1
0	0	0	0	1	1	0	1

To obtain 13, we put 1 under 8, 4, and 1 as the sum of these numbers is equal to 13, and the remaining bits will be zero. Therefore, the binary bit version of 13 is 00001101.

What is a Public IP Address?

A public IP address is an address provided that is provided by your internet service provider (ISP) to your network. On the internet, it recognizes your device. The internet is accessed through your router's public IP address. Public IP addresses are commonly used by publicly accessible enterprises such as websites, DNS, and VPN servers because they can be accessed from anywhere in the world. You can not go online without having your public IP address, which identifies your device on the internet. Your router is an intermediate between your computer and the internet on a typical home network.

What is a Private IP Address?

A private IP address, also known as the local IP address, is the IP address your network router allocates to your device. This address is only visible within your network, so it is unavailable on the Internet. Each device on the same network is assigned a unique private IP address that allows them to communicate with other devices. The device in your home can have the same private IP address as your neighbors' device or anyone else's all over the world, with private IP addresses. This is due to the non-routable nature of private addresses.

Key Differences Between Public and Private IP Addresses

The following are the key differences between a private and a public IP address.

Sr. No	Key	Private IP Address	Public IP Address
1.	Definition	A private IP address, also known as the local IP address, is the IP address that your network router allocates to your device.	A public IP address is the public-facing IP address provided to your network by your internet service provider (ISP).
2.	Scope	These addresses have local scope. Devices within a network use this to communicate.	These addresses have a global scope.
3.	Communicatio n	A private IP address is usually used to communicate with any other device in your house or office that is connected to your private network.	A public IP address is usually used to communicate outside your network.
4.	Format	Private IP Addresses differ uniformly.	Public IP Addresses differ in varying ranges.
5.	Traffic	All data transfers using a private IP address stay within the network and are unaffected by external factors.	The Internet Service Provider (ISP) control s the data transfers using a public IP address.
6.	Provider	A Local Network Provider creates the private IP addresses by the use of network operating systems.	The public IP addresses are controlled by Internet service providers (ISP).
7.	Cost	Private IP Addresses are free of cost.	Public IP Address comes with a cost.
8.	Locate	Private IP Address can be found by entering "ipconfig" on the command prompt.	Public IP Address needs to be searched on search engines like google.
9.	Numeric code	Private IP addresses use numeric code that is not unique and can be reused by any device connected to a private network.	Public IP addresses use a numeric code that is unique to each device and cannot be repeated.
10.	Security	Private IP addresses are secure.	Public IP addresses are vulnerable to attack since they have no security.

Sr. No	Key	Private IP Address	Public IP Address
11.	Range	The private IP addresses range from: 10.0.0.0 to 10.255.255.255, 172.16.0.0 to 172.31.255.255, and 192.168.0.0 to 192.168.255.255	Except for private IP addresses, all other IP addresses are public IP.
12.	Example	192.168.11.20 is an example of a private IP address.	17.5.7.3 is an example of a public IP address.

OUTPUT SCREENSHOT:

- *STICK HERE YOUR OUTPUT IMAGES*

DO PROPER TAGGING

OBSERVATION:

- *TYPE HERE*

CONCLUSION: Hence we understand the concept of networking to configure the Router, Access point, and IPV4 for private and public IPs.

DISCUSSION QUESTIONS:

1. Explain Public IP?
2. What is Private IP?
3. What is IPV4?
4. Explain Networking between two or more computers.

REFERENCE:

- https://support.microsoft.com/en-us/windows/file-sharing-over-a-network-in-windows-b58704b2-f53a-4b82-7bc1-80f9994725bf
- https://www.techtarget.com/searchenterprisedesktop/tip/Device-drivers-Installation-and-configuration
- https://www.scaler.com/topics/computer-networsk/public-and-private-ip-address/
- https://www.cisco.com/c/en/us/solutions/small-business/resource-center/networking/how-to-set-up-router.html#~router-setup-steps

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Activity No. 7

Aim: Demonstrate the concept of RDP for remotely controlling another computer from our system learn other remote desktop solutions and learn the data security and data sharing concepts.

Name of Student:

Roll No.:

Semester/Year:

Academic Session:

Date of Performance:

Date of Submission:

AIM: Demonstrate the concept of RDP for remotely controlling another computer from our system learn other remote desktop solutions and learn the data security and data sharing concepts.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

Understand the concept of RDP for remotely controlling another computer from our system, learn other remote desktop solutions, and learn the data security and data sharing concepts.

HARDWARE AND SOFTWARE REQUIREMENTS:

Hardware Requirement

- Operating systems: Windows 10 1703 or later.
- CPU: 1 GHz or faster processor.
- RAM: 1024 MB.
- Hard drive: 100 MB or more.
- Video: DirectX 9 or later with WDDM 1.0 driver.

THEORY:

STEPS FOR PRACTICAL:

- **What is the Remote Desktop Protocol (RDP)?**

  The Remote Desktop Protocol (RDP) is a protocol, or technical standard, for using a desktop computer remotely. Remote desktop software can use several different protocols, including RDP, Independent Computing Architecture (ICA), and virtual network computing (VNC), but RDP is the most used protocol. RDP was initially released by Microsoft and is available for most Windows operating systems, but it can be used with Mac operating systems too.

- **What does 'remote desktop' mean?**

  A remote desktop is the ability to connect with and use a faraway desktop computer from a separate computer. Remote desktop users can access their desktop, open and edit files, and use applications as if they were sitting at their desktop computer. Employees often use remote desktop software to access their work computers when they are traveling or working from home.

- Remote desktop access is very different from cloud computing, even though both allow employees to work remotely. In cloud computing, users access files and applications that

are stored in the cloud — specifically, in cloud servers. In contrast, when using remote desktop software, users are accessing their physical desktop computer, and can only use files and applications saved locally on that desktop. Cloud computing is sometimes easier to use and more efficient to implement for remote workforces, but many companies have not migrated to the cloud, or cannot for security or regulatory reasons.

- **How does RDP work?**

  Think of a remote-controlled drone or toy car. The user presses buttons and steers the drone or car from afar, and their commands are transmitted to the vehicle. Using RDP is somewhat like that: the user's mouse movements and keystrokes are transmitted to their desktop computer remotely, but over the Internet instead of over radio waves. The user's desktop is displayed on the computer they are connecting to, just as if they were sitting in front of it.

- The RDP protocol opens a dedicated network channel for sending data back and forth between the connected machines (the remote desktop and the computer currently in use). It always uses network port 3389 for this purpose. Mouse movements, keystrokes, the desktop display, and all other necessary data are sent over this channel via TCP/IP, which is the transport protocol used for most types of Internet traffic. RDP also encrypts all data so that connections over the public Internet are more secure.

- Because keyboard and mouse activity have to be encrypted and transmitted over the Internet, which takes a few milliseconds, and because the desktop display has to be transmitted back to the user, slight delays often occur. For instance, if a user double-clicks on an application to open it, the "double click" may not take place for a few milliseconds as the user's action is transmitted to the desktop before being carried out. Then, when the application opens, there may be another short delay while the display is transmitted back to the user.

- **What are the pros and cons of using RDP?**

  There are several advantages to the RDP. One advantage is that it does not require a VPN. It also keeps data stored securely on the user's desktop, instead of storing it on cloud servers or the user's unsecured personal devices. Furthermore, RDP enables companies

with a legacy on-premises IT setup to allow their employees to work from home.

However, RDP may cause users to experience lag, especially if their local Internet connection is slow. This can frustrate remote employees and cut down on their productivity. RDP also has some serious security vulnerabilities that leave it open to cyber attacks. (Learn more about RDP security.)

Steps to Secure Your RDP

1. Rigorous Patch Management:

Continually update both the operating system and applications. This includes subscribing to updated services and conducting regular audits to ensure all systems are up-to-date.

2. Multi-Factor Authentication (MFA):

Beyond two-factor, consider implementing multi-factor authentication. This approach combines something you know (password), something you have (security token), and something you are (biometrics), significantly increasing security.

3. Restrict Privileged Access:

Implement the principle of least privilege by ensuring users have only the access necessary for their role. Review and adjust these permissions regularly to adapt to changing roles and responsibilities.

4. Strong, Unique Passwords

Enforce the use of strong, unique passwords for RDP access. Employ password managers to generate and store complex passwords and mandate regular password changes.

5. Secure RDP Files

Avoid storing passwords in RDP files and disable the option to save credentials as a default setting on client devices. This prevents credential theft of lost or stolen devices.

6. Manage RDP Connection History

Regularly clean the RDP connection cache to remove residual information that could be used in an attack. Automate this process through scripts or group policies where possible

7. Network Level Authentication (NLA)

Require NLA for RDP sessions, which necessitates that the connecting user authenticate before a session is established, thus providing an additional layer of security against unauthorized access.

8. Configure Firewalls Strictly

Limit RDP access through firewalls by whitelisting IP addresses. Employ additional monitoring to detect and respond to unusual access patterns or unauthorized remote access or attempts.

9. Deploy RDP Gateways

Use Remote Desktop Gateway servers as a middle layer to manage and secure RDP traffic entering the corporate network from remote systems. RD Gateways uses SSL encryption, improving security over traditional RDP.

10. Alter Default RDP Port

- Change the default port (3389) to a less predictable number to avoid targeted attacks on this well-known port. Monitor and potentially rotate the RDP port periodically to enhance security.

OUTPUT SCREENSHOT:

- *STICK HERE YOUR OUTPUT IMAGES*

DO PROPER TAGGING

OBSERVATION:

- *TYPE HERE*

CONCLUSION: Hence, we understand the concept of RDP for remotely controlling another computer from our system learn other remote desktop solutions and learn the data security and data sharing concepts

- **DISCUSSION QUESTIONS:**
  1. What is the Remote Desktop Protocol (RD)
  2. What does 'remote desktop' mean?
  3. How does RDP work?
  4. What are the pros and cons of using RDP?

REFERENCE:

- https://www.google.com/search?q=software+and+hardware+requirements+for+rdp&oq=&gs_lcrp=EgZjaHJvbWUqCQgGECMYJxjqAjIJCAAQIxgnGOoCMgkIARAjGCcY6gIyCQgCECMYJxjqAjIJCAMQIxgnGOoCMgkIBBAjGCcY6gIyCQgFECMYJxjqAjIJCAYQIxgnGOoCMgwIBxAjGCcY6gIYiwPSAQk2NzY4ajFqMTWoAgiwAgE&sourceid=chrome&ie=UTF-8
- https://www.cloudflare.com/learning/access-management/what-is-the-remote-desktop-protocol/